

This problem set will not be marked. Nevertheless, it is probably in your interest to work on them.

----- ★ -----

1. Inertial frame S' moves at velocity v with respect to another inertial frame S , in the common x direction. The two frames are coincident at time $t = t' = 0$.

- (a) [*** pts**] Consider two events that occur at positions (x_1, y_1, z_1) and (x_2, y_2, z_2) , and at times t_1 and t_2 . Write down the Lorentz transformations for both events. Hence deduce the transformations for the interval between them, $\Delta \mathbf{r} = (\Delta x, \Delta y, \Delta z)$ and Δt . Do the differences $\Delta \mathbf{r}$ and Δt transform exactly the same way as $\mathbf{r}$ and t ?
- (b) [*** pts**] An object moves with constant velocity $\vec{u}'$ (components u'_x , u'_y , and u'_z) relative to S' . (If the particle changes x -position by $\Delta x'$ in time interval $\Delta t'$ as observed in the S' frame, then $u'_x = \Delta x' / \Delta t'$.) Use the (inverse) Lorentz transformation to show that the velocity components (u_x, u_y, u_z) of the object relative to the S frame are

$$u_x = \frac{u'_x + v}{1 + u'_x v / c^2}, \quad u_y = \frac{u'_y}{\gamma(v) (1 + u'_x v / c^2)}, \quad u_z = \frac{u'_z}{\gamma(v) (1 + u'_x v / c^2)}.$$

2. Four-momenta.

- (a) [*** pts**] Show that the four-momentum of a particle with nonzero mass is always timelike.
- (b) [*** pts**] Write down the four-momentum of a photon moving in the y direction. The frequency of the photon is denoted by f .

3. [*** pts**] A neutral pion has mass M and is traveling with speed v when it decays into two photons. The photons are seen to emerge at equal angles θ on either side of the original velocity. Show that $v = c \cos \theta$.

(As usual, with decay/reaction problems, draw clear before & after pictures, write equations for conservation of energy and momentum, choosing the variables more suitable for the problem [express in terms of speeds and γ 's, or in terms of momenta?], and then work with those equations.)

4. [*** pts**] When he returns his rented rocket after one week's cruising in the galaxy, Mr. Spock is shocked to be billed for three weeks' rental. Assuming that he traveled straight out and straight back, always at the same speed, how fast was he traveling?

5. [*** pts**] Use the fact that rapidities are additive to derive the relativistic formula for addition of velocities along the same direction.

In other words, if frame S' has speed $v_1 = c \tanh \phi_1$ relative to frame S , and frame S'' has speed $v_2 = c \tanh \phi_2$ relative to frame S' (in the same direction), use the fact that the rapidity of S'' relative to S is $(\phi_1 + \phi_2)$, to find the speed of S'' relative to S .

You may need to derive an identity concerning the hyperbolic tangent of the sum of two quantities. You can derive such an identity using the relations

$$\sinh(z_1 + z_2) = \sinh z_1 \cosh z_2 + \cosh z_1 \sinh z_2$$

$$\cosh(z_1 + z_2) = \cosh z_1 \cosh z_2 + \sinh z_1 \sinh z_2$$

and the definition $\tanh z = \sinh z / \cosh z$.

6. Four-velocities.

- (a) [*** pts**] Write down the four-velocity of a particle with nonzero mass m moving in the y direction with speed v . Show that this four-vector is timelike.
- (b) [*** pts**] Write down the four-velocity of a particle with nonzero mass m and velocity components v_x , v_y , v_z . Show that this four-vector is timelike.
- (c) [*** pts**] What is the four-velocity of the particle with mass m in its own rest frame, i.e., in the frame where it is at rest?
- (d) [*** pts**] Try writing down the four-velocity of a photon moving in the y direction. Is there a problem?

7. [*** pts**] What is a particle's speed if its kinetic energy T is equal to its rest energy? What is a particle's speed if its kinetic energy is n times its rest energy?

(Reminder: the kinetic energy is the total energy minus the rest energy.)

8. Inertial frame S' moves at velocity v with respect to another inertial frame S , in the common x direction. A particle has velocity in the common x direction. Its speed is u in the S frame and u' in the S' frame.

- (a) [*** pts**] Write down the four-velocity of the particle in the S frame and in the S' frame.
- (b) [*** pts**] Since the four-velocity is a four-vector, it should transform according to a Lorentz transformation. Write down the transformations between the four-vector components. Be careful to indicate the relevant speed for each γ .
- (c) [*** pts**] Use the transformation equations to obtain u' in terms of u .
- (d) [*** pts**] Now write down the inverse transformations, and use these to obtain u in terms of u' .

Comment: You should obtain the velocity addition formula through this procedure.

9. [*** pts**] Two balls of equal mass (m each) approach each other with equal but opposite velocities of magnitude $v = 0.8c$. Their collision is perfectly elastic, so they stick together and form a single body of mass M . What is the velocity of the final body and what is its mass M ?

Please provide answers first in terms of the symbol v , and only at the end use the value $v = 0.8c$.

10. Velocity addition:

- (a) [*** pts**] A rocket traveling at speed $\frac{1}{2}c$ with respect to frame S shoots forward bullets at speed $\frac{3}{4}c$ relative to the rocket. What is the speed of the bullets relative to S ?
- (b) [*** pts**] If $w = (u + v)/(1 + uv/c^2)$, derive a relation of the form

$$\gamma(w) = \gamma(u)\gamma(v) \times \text{something}$$

and derive what this something is. Do this first using $c = 1$ units (less chance of making mistakes). Then perform the same derivation while retaining the c 's.

- (c) [*** pts**] Rocket B moves at speed u directly away from earth. It emits a bullet at speed v perpendicular to the direction of its motion. Measured with respect to earth, what are the velocity components of the bullet? Measured with respect to earth, what is the speed of the bullet?